

Introduction

Data

The dataset used in this coursework is the **Video Game Sales & Industry Data (1980 - 2024)**, sourced from Kaggle and available at: <https://www.kaggle.com/datasets/asaniczka/video-game-sales-2024>. It contains 64,000+ records, with each row representing one game-platform combination. The dataset contains global as well as regional sales along with critic review scores. The dataset was available in CSV format, was downloaded, and stored locally. This dataset was selected because it was publicly available and clearly documented and contained enough categorical and quantitative variables to support analysis. The file size was well within Tableau's recommended performance guidelines.

To support and check the quality of the data, three variables were derived during data preparation. These variables are described in the data dictionary below and are highlighted in **bold**:

Name	Description	Domain
Title	The commercial title of the video game.	Nominal
Console	The gaming platform or console for which the game was released.	Nominal
Genre	The genre classification of the game, used to group titles by type of gameplay.	Nominal
Publisher	The company which published and distributed the game.	Nominal
Developer	The organization that developed the game.	Nominal
Critic_Score	The overall critic review score sourced from Metacritic.	Float [0.0–10.0]
Total_Sales	Total global sales of the game across all the regions. Measured in millions of units sold.	Float ≥ 0 , in millions
NA_Sales	Sales in North American market, in millions of units sold.	Float ≥ 0 , in millions
JP_Sales	Sales in Japanese market, in millions of units sold.	Float ≥ 0 , in millions
PAL_Sales	Sales in the PAL territory (Europe and Africa), in millions of units.	Float ≥ 0 , in millions
Other_Sales	Sales in all remaining markets not captured by the above regional columns, in millions of units.	Float ≥ 0 , in millions
Release_Date	The date of commercial release for the game on the specified platform.	[1980 – 2024]
Check_Total [Derived]	Calculated as the sum of NA_Sales + JP_Sales + PAL_Sales + Other_Sales.	Float ≥ 0 , in millions

Data Quality Note: It should be noted that Critic_Score field contains significant number of null values, as many old games do not have any entries from Metacritic. For the analysis of certain features, records with null values will be excluded via a filter. All other variables used are complete across the dataset.

Persona and questions

The user is an investor within the business development sector and is trying to identify high-return opportunities in the gaming market across all regions. They have three simple and one complex question:

1. Which video game genres generate the highest total global sales, and how are those sales distributed across regional markets? **[simple question – SQ1]**
2. Is there a positive relationship between a game's critical reception and its total global sales? **[simple question – SQ2]**
3. How have global video game sales evolved across decades, and which genres have grown or declined over time? **[simple question – SQ3]**
4. Among the top-selling genres, which specific publisher-console combinations drive regional dominance, and does critical score correlate with commercial success within those combinations? **[complex question – CQ]**

Requirements

- **SQ1-R1** – The user needs to visualise Genre (categorical) in relation to Total_Sales and regional measures (NA_Sales, JP_Sales, PAL_Sales, Other_Sales) to convey both the ranking of genres and the part-whole regional breakdown. A stacked horizontal bar chart with Genre on the y-axis and Total_Sales on the x-axis, with four colour-coded regional segments per bar, sorted descending by total length. The longest bar identifies the highest-grossing genre; relative segment widths reveal regional market concentration (Mackinlay, 1986).
- **SQ2-R2** – The user needs to visualise the correlation between Critic_Score (x-axis) and Total_Sales (y-axis). A scatter plot with each mark representing one game-platform record, overlaid with a linear regression trend line and confidence band. Null Critic_Score records excluded. Mark opacity set to 30% to address overplotting. An upward-sloping trend line indicates positive correlation; a flat line indicates weak predictive power (Bertin, 1983).
- **SQ3-R3** – The user needs to visualise Release_Date (year), Total_Sales, and Genre to convey time-series trends across decades. A multi-series line chart with year on the x-axis, Total_Sales on the y-axis, and one coloured line per top five genre. Rising lines indicate growing genres; falling lines indicate decline; crossing lines show genre displacement (Few, 2012).
- **CQ-R4** – The user needs to investigate Genre, Publisher, Console, regional sales measures, and Critic_Score simultaneously-five or more variables that cannot be encoded in a single static view. The SQ1 bar chart acts as the primary control: clicking a genre bar triggers a Tableau Dashboard Action that filters a Publisher-Console Heatmap (colour intensity = Selected Region Sales parameter, switchable between NA/JP/PAL). The analyst reads the heatmap to identify the darkest publisher-console cell in their target region, then inspects the filtered view to assess critic-commercial correlation within that genre (Shneiderman, 1996).

Non-Functional Requirements

- **NFR1 – Hardware and Display:** The dashboard must display correctly within 1920×1080 pixels with no scrolling required.
- **NFR2 – Context of Use:** Intended for indoor professional desktop use with mouse-based interaction.
- **NFR3 – Visual Literacy:** The persona has intermediate visual literacy; all charts must include clear axis labels, titles, and legends.

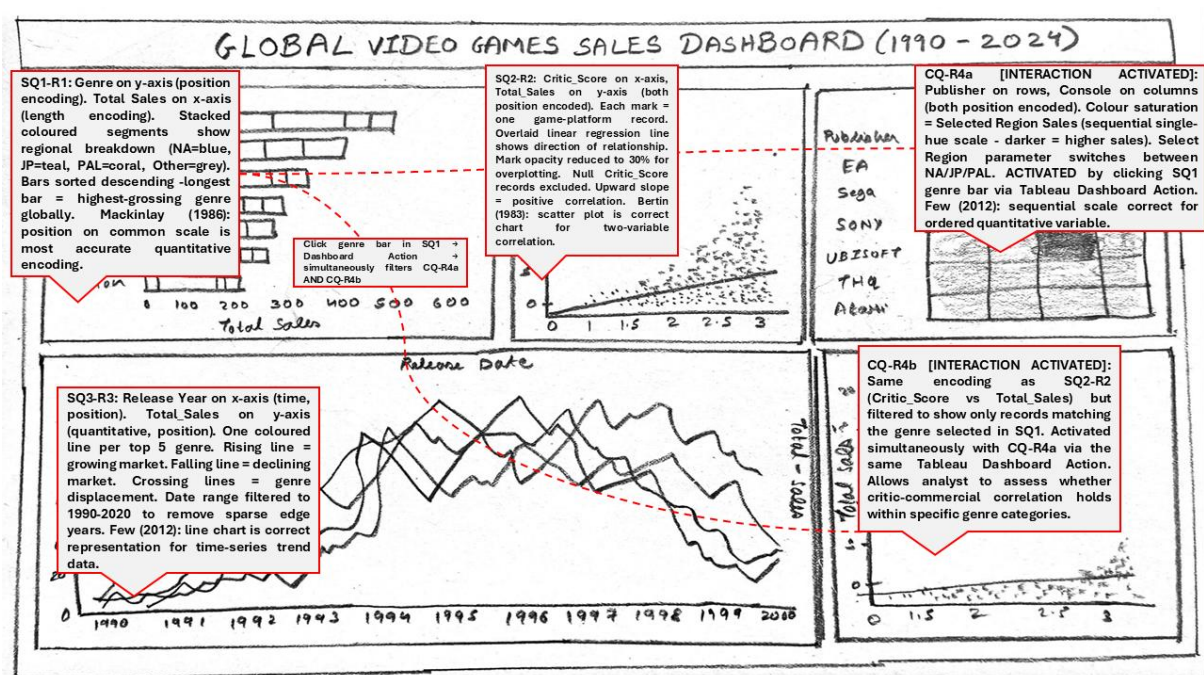
- **NFR4– Aesthetics:** A clean professional colour scheme must be maintained; colour encodes data only and is not used decoratively.

Cross-Reference Summary

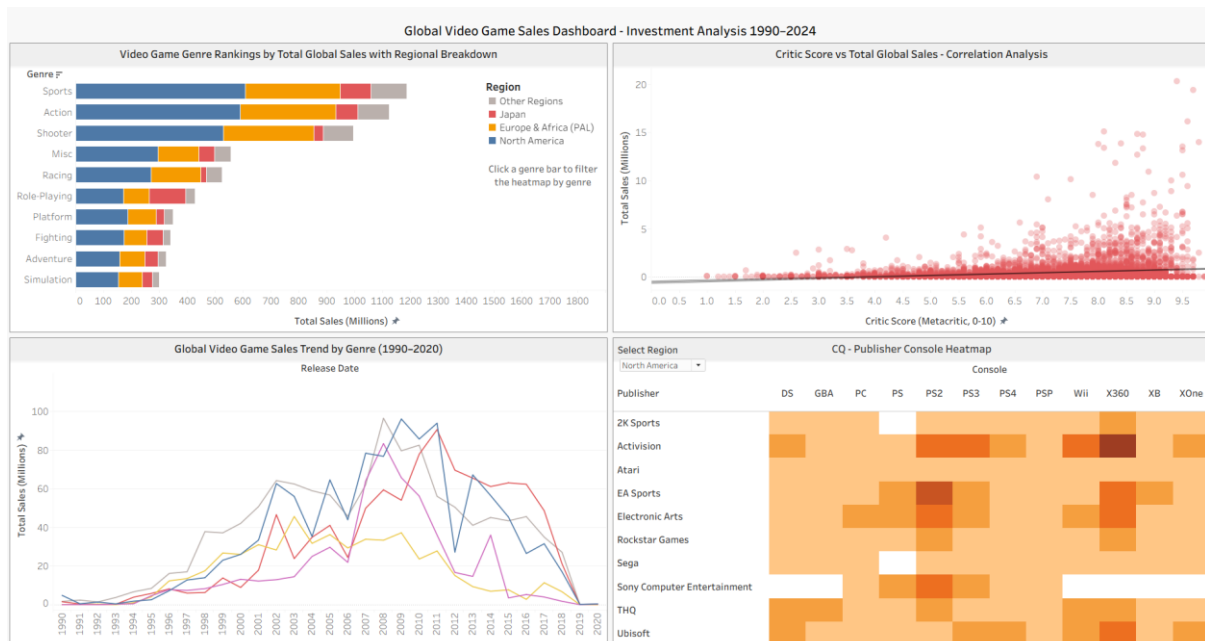
Requirement	Question	Chart Type
SQ1-R1	Genre ranking and regional breakdown.	Stacked horizontal bar chart.
SQ2-R2	Critic score vs total sales correlation.	Scatter plot with regression trend line.
SQ3-R3	Sales trend over time by genre.	Multi-series line chart.
CQ-R4	Genre, publisher, console, region, and critic score.	Heatmap + scatter (coordinated via Dashboard Actions).
NFR1-NFR4	All questions	Dashboard-level (display, context, literacy, aesthetics).

Design

2.1 Paper Landscape Design: Before building anything in Tableau, a paper landscape design was created to plan how the dashboard would look and function. This approach, introduced in the Week 24/25 lab sessions, meant that all design decisions were made based on visualisation theory rather than being shaped by what Tableau can or cannot do. The design proposed five views arranged across a 1920×1080 frame: a stacked horizontal bar chart for SQ1 in the top-left, a scatter plot for SQ2 in the top-right, a multi-series line chart for SQ3 spanning the bottom-left, and two interaction-activated views for the complex question - a publisher-console heatmap (CQ-R4a) and a genre-filtered scatter plot (CQ-R4b) - positioned in the right column. Both CQ views were marked with dashed arrows in the sketch to make clear that they only populate when the user interacts with the dashboard rather than being statically shown. Each view is accompanied by an annotation callout box describing the visual encoding choices and the theoretical justification for each design decision, cross-referenced to the relevant requirement.



The implemented dashboard is shown below. The core layout and chart type choices remain consistent with the paper landscape, but four changes were made during the implementation process.



Change 1 - Five views reduced to four. The paper landscape proposed a fifth view, CQ-R4b, which was a genre-filtered version of the SQ2 scatter plot activated alongside the CQ heatmap. During implementation, fitting five views within the 1920×1080 boundary without scrolling forced every chart to a size where axis labels and data marks became noticeably harder to read. On top of that, the filtered scatter turned out to be redundant - the heatmap combined with the Dashboard Action and the Select Region parameter already answered the complex question fully without needing an extra view. Removing it also aligned with the module advisor's guidance that four views is preferable unless a fifth clearly adds something new. The result is a cleaner dashboard where each chart has enough space to be properly readable, which is more important than including every originally planned element.

Change 2 - Top ten genre filter on SQ1. The paper landscape did not specify a genre limit. When all genres were displayed in Tableau, the chart became overcrowded with approximately twenty bars, many of which had negligible sales and added visual noise without contributing any useful information. Applying a Top 10 filter by total sales kept the commercially relevant genres while making the chart significantly easier to read.

Change 3 - Date range restricted to 1990–2020 on SQ3. This was not planned in the paper landscape. During implementation, years before 1990 produced near-flat lines because the dataset has very few records from that period, and years after 2020 showed a sharp artificial decline that reflects incomplete data collection for recent releases rather than any genuine market contraction. Restricting the range to 1990–2020 produces a more honest and readable representation of the actual market trends.

Implementation

The dashboard was implemented using Tableau Desktop. The original design proposed five views; as described in the Design section, this was reduced to four during implementation following readability and redundancy considerations. The four implemented sheets are: SQ1 - Genre Sales by Region (stacked horizontal bar chart), SQ2 - Critic Score vs Total Sales (scatter plot), SQ3 - Sales Trend Over Time (multi-series line chart), and CQ - Publisher Console Heatmap. The dataset was connected as a local CSV file, and field types were verified in the Data Source tab before any sheets were built.

SQ1 - Genre Sales by Region (Stacked Horizontal Bar Chart). Genre was placed on Rows, Measure Values on Columns, and Measure Names on Colour to create the stacked bar structure: NA_Sales, JP_Sales, PAL_Sales, and Other_Sales. Bars were sorted in descending order of total sales by right clicking the Genre axis and selecting Sort by Field (Total_Sales, Sum, Descending). A Top 10 filter was applied to Genre to retain only the ten highest-grossing genres. The default Tableau color palette was replaced with a custom scheme: NA_Sales = steel blue, JP_Sales = red, PAL_Sales = gold/yellow, Other_Sales = grey. Legend aliases were edited to display descriptive region names rather than field names. Gridlines were removed via Format

→ Lines. The chart title was set to "Video Game Genre Rankings by Total Global Sales with Regional Breakdown" and the x-axis label to "Total Sales (Millions)".

SQ2 - Critic Score vs Total Sales (Scatter Plot). Critic_Score was placed on the Columns shelf and Total_Sales on the Rows shelf. Aggregate Measures was disabled via the Analysis menu to plot individual game records rather than aggregated totals. A filter was applied to Critic_Score to exclude null values, as many older and smaller titles have no Metacritic entry. Mark opacity was set to 30% via the Color card to mitigate severe overplotting across 64,000+ records. A linear regression trend line was added via the Analytics pane with confidence bands enabled. The R^2 value was noted from the trend line descriptor for inclusion in the Walkthrough section.

SQ3 - Sales Trend Over Time (Multi-Series Line Chart). YEAR(Release_Date) was placed on Columns and Total_Sales on Rows. Genre was dragged to the Color card to produce one colored line per genre. A Top 5 filter was applied to Genre by total sales to limit the view to the five highest-grossing genres, preventing visual clutter from the approximately twenty genre categories in the dataset. A date range filter was applied restricting the view to 1990–2020. The mark type was set to Line. A custom categorical color palette was applied using maximally distinct hues to ensure all five lines are clearly distinguishable.

CQ - Publisher Console Heatmap. Publisher was placed on the Rows shelf and Console on the Columns shelf. Top ten filters were applied to both Publisher and Console. A calculated field named Selected Region Sales was created using an IF/ELSEIF formula referencing a string parameter named Select Region. This parameter allows the analyst to switch the heatmap color encoding between NA_Sales, JP_Sales, PAL_Sales, and Total_Sales. The Selected Region Sales calculated field was placed on the Color card. A sequential single-hue orange color scale was applied. The mark type was set to Square.

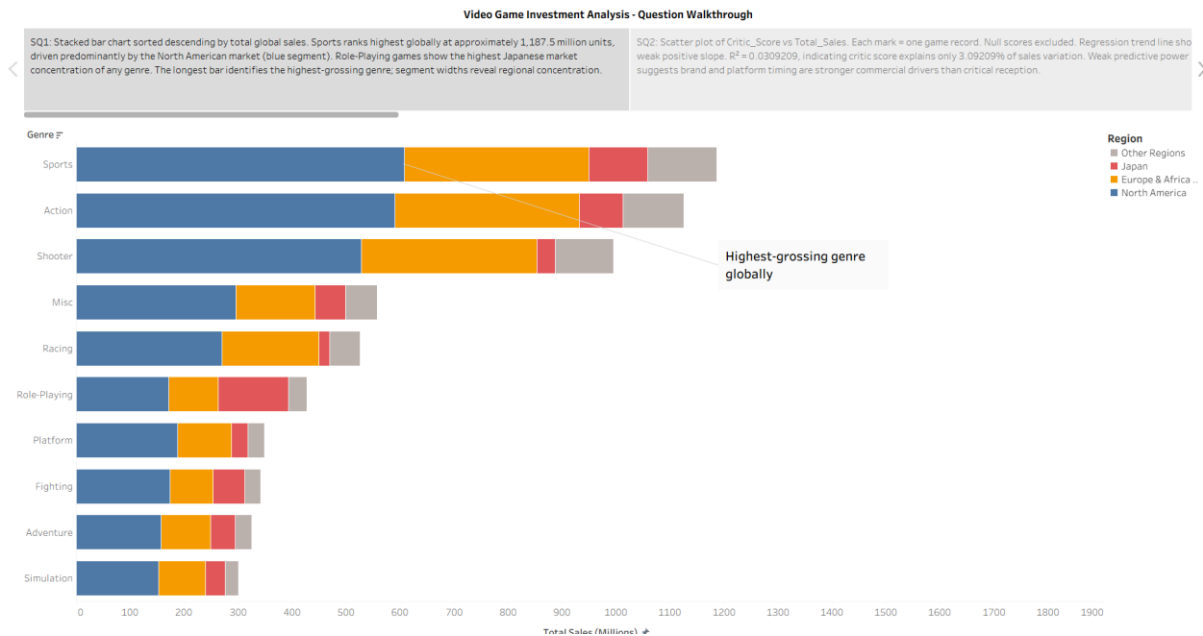
Dashboard Assembly. A new dashboard was created at 1920×1080 fixed size. All four sheets were arranged in a 2×2 tiled layout. Each sheet was set to Entire View fit. A white background was applied to the dashboard canvas via the Layout panel to create visual separation between chart panels. A Dashboard Filter Action named Genre Selection was configured with SQ1 as the source sheet and the CQ heatmap as the target, running on Select with Show All Values on clearing. The Select Region parameter control was positioned above the CQ heatmap. A text instruction was added to the SQ1 panel directing the analyst to click a genre bar to filter the heatmap. A dashboard title was added at the top.

Story Sheet. A Tableau Story sheet was created with the title "Video Game Investment Analysis - Question Walkthrough". Six story points were created: three for the simple questions (one per SQ), and three for the complex question showing the initial dashboard state, the post-genre-selection filtered heatmap state, and the post-region-switch state. Each story point includes an annotated caption and one or more in-context mark annotations highlighting key findings.

Walkthrough

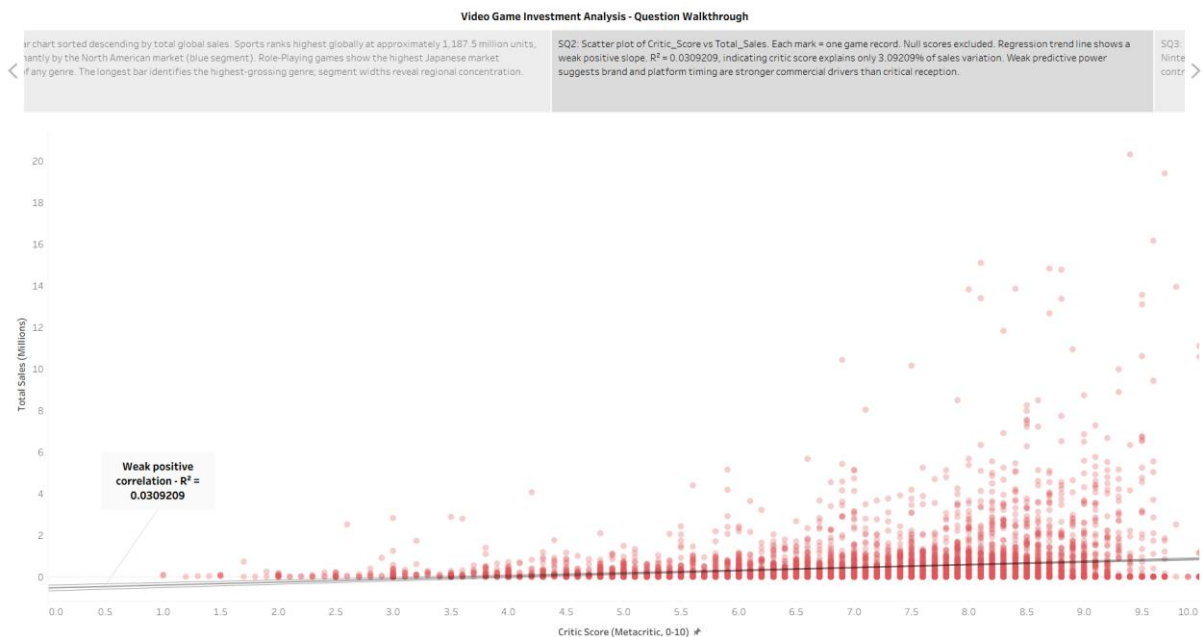
The following section walks through how each question was answered using the dashboard. Screenshots from the Tableau Story sheet are referenced throughout. The Story sheet is titled "Video Game Investment Analysis-Question Walkthrough" and is included in the submitted packaged workbook.

SQ1-Which video game genres generate the highest total global sales, and how are those sales distributed across regional markets?



As shown in Story Point 1, the stacked bar chart displays the top ten genres sorted from highest to lowest total global sales. Sports sit at the top with approximately 1,187.5 million units sold globally, followed closely by Action at approximately 1,125.9 million. Looking at the colored segments within each bar, it is immediately clear that both top genres are heavily driven by the North American market-the blue segment accounts for most total sales in both cases. Role-Playing games tell a different story: although they rank 6 overall, their regional profile is noticeably different, with the Japanese segment (red) making up a much larger share compared to any other genre. The answer to SQ1 is therefore that Sports and Action dominate global sales, but the regional composition of those sales varies considerably across genres, which has direct implications for where a publisher chooses to focus their market strategy.

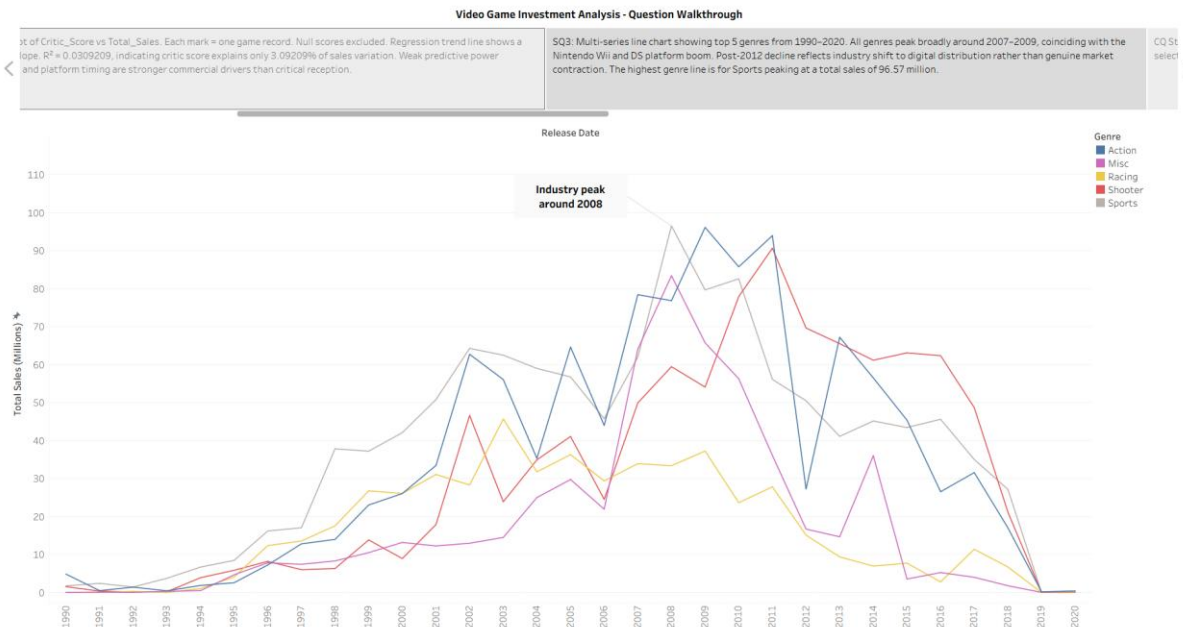
SQ2-Is there a positive relationship between a game's critical reception and its total global sales?



Story Point 2 shows the scatter plot of Critic_Score against Total_Sales across all game records with a valid Metacritic entry. The regression trend line has a shallow upward slope, suggesting a weak positive relationship-games with higher critic scores do tend to sell slightly more on average. However, the R^2 value of 0.0309209 tells a more important story: critic score only explains 3.09209% of the variation in sales figures, meaning the vast majority of what determines a game's commercial performance lies elsewhere.

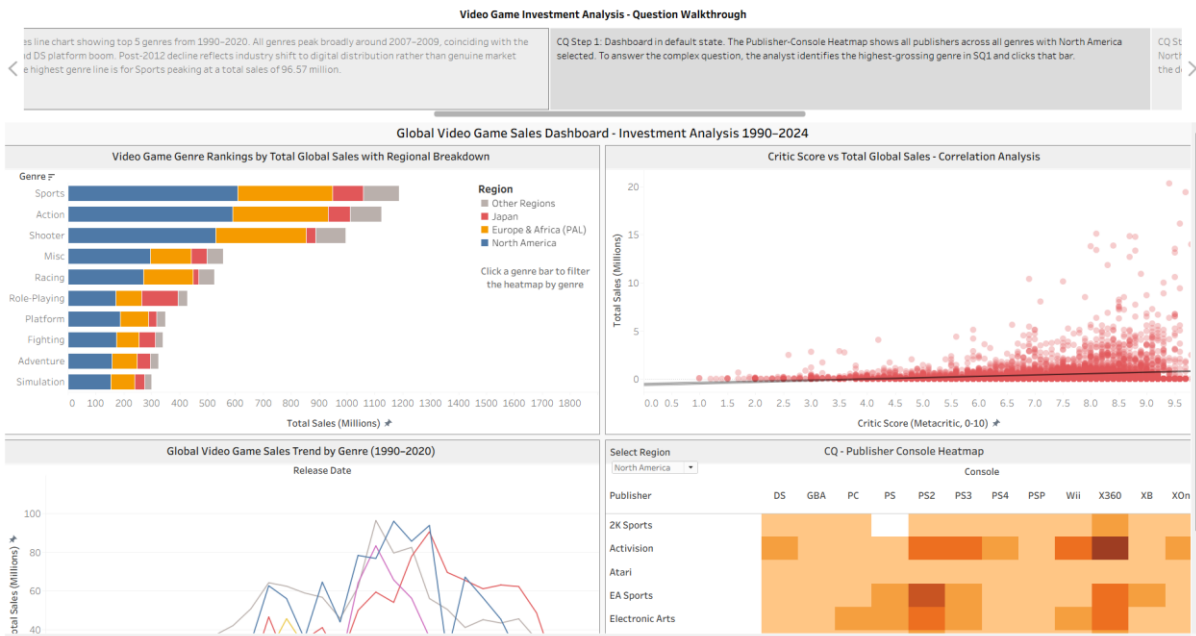
There are also notable marks in the bottom-right area of the chart games with low critic scores but strong commercial performance—which are likely franchise titles where brand recognition outweighed review scores entirely. It suggests that a publishing strategy built primarily on backing critically praised titles would be unreliable, and that factors like IP strength, platform timing, and marketing reach are likely much stronger commercial drivers.

SQ3-How have global video game sales evolved over decades, and which genres have grown or declined over time?

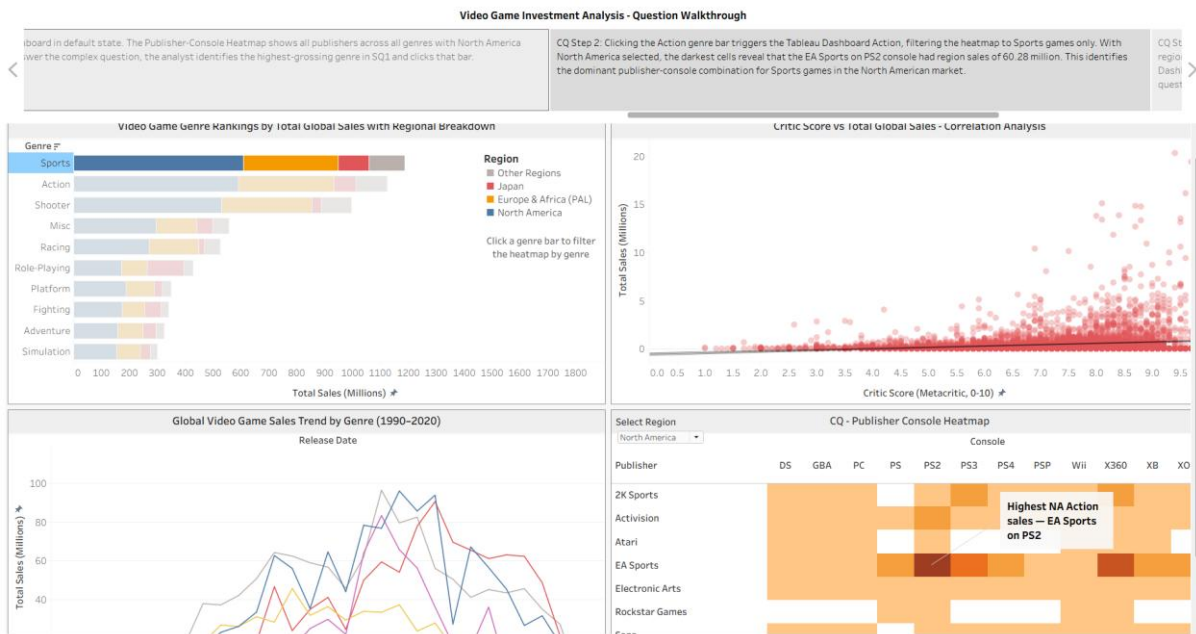


Story Point 3 shows the multi-series line chart covering 1990 to 2020. All five genre lines rise steadily from the early 1990s, peak broadly around 2007 to 2009, and then decline through the 2010s. The 2007–2009 peak coincides with the commercial explosion of the Nintendo Wii and DS platforms, which brought gaming to a much wider audience across multiple genres simultaneously. The post-2010 decline more likely reflects the industry's shift toward digital distribution than genuine market contraction.

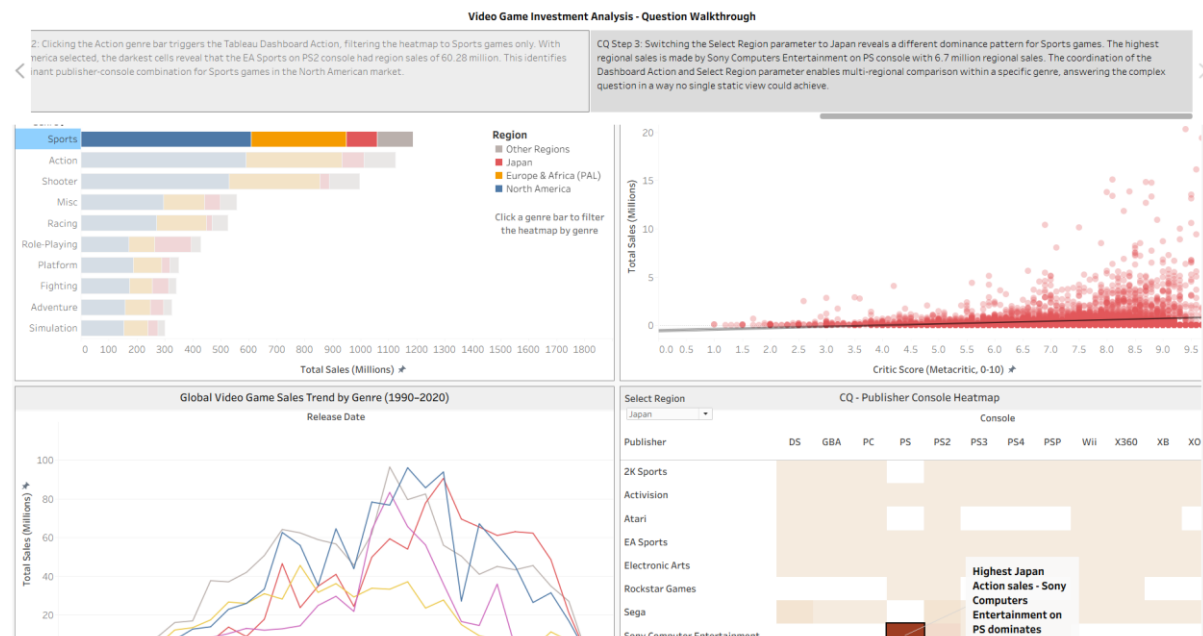
CQ-Among the top-selling genres, which specific publisher-console combinations drive regional dominance?



Story Points 4, 5, and 6 demonstrate the interactive walkthrough for the complex question. In Story Point 4, the dashboard is shown in its default state with no genre selected-the heatmap displays all publishers across all genres with North America as the default region, giving the analyst an overview of the full market structure before narrowing down.



In Story Point 5, the Action genre bar in SQ1 has been clicked. The Tableau Dashboard Action fires immediately, filtering the heatmap to show only Action genre data. With North America selected via the region parameter, the darkest cells in the heatmap reveal that EA Sports on PS2 had highest regional sales. This directly answers the first part of the complex question.



Story Point 6 shows what happens when the Select Region parameter is switched to Japan with Action still selected. The color pattern in the heatmap shifts noticeably highest Japan sales was for Sony Computers Entertainment on PS console. Switching to PAL reveals a third distinct pattern for the European market. The key insight from this interaction is that publisher-console dominance is not consistent across regions—the combinations that perform best in North America are not the same as those that lead in Japan or Europe. For the investor persona, this means that regional market entry decisions should be made based on region-specific publisher data rather than global aggregate figures alone. The coordination between the SQ1 bar chart, the Dashboard Action, and the Select Region parameter makes this kind of multi-dimensional regional analysis possible in a way that no single static view could achieve.

Conclusion

This project developed an interactive Tableau dashboard to support video game investment decision-making, using a dataset of 64,000+ game-platform records spanning 1980–2024. Three simple questions and one complex question were addressed through a four-view dashboard comprising a stacked bar chart, scatter plot, multi-series line chart, and publisher-console heatmap. Key findings were that Sports and Action dominate global sales with strong North American concentration, that critic score is a weak commercial predictor ($R^2=0.03$), that sales peaked around 2007–2009 reflecting the Wii and DS platform boom, and that publisher-console regional dominance patterns vary significantly across NA, JP, and PAL markets. View coordination through a Tableau Dashboard Action and Select Region parameter enabled the complex question to be answered interactively in a way no single static view could achieve.

References

- Bertin, J. and Berg, W.J. (2011) *Semiology of graphics : diagrams, networks, maps*. 1st ed. Redlands, Calif. :, [LaVergne, Tenn.] : Esri Press ; Distributed by Ingram Publisher Services.
- Osińska, V. (2015) 'Alberto Cairo: *The Functional Art. An Introduction to Information Graphics and Visualization*, Berkeley 2012', *Toruńskie Studia Bibliologiczne*, 7, p. 207. <https://doi.org/10.12775/TSB.2014.025>
- Few, S. (2012) *Show Me the Numbers: Designing Tables and Graphs to Enlighten*. 2nd edn. Burlingame, CA: Analytics Press.
- Mackinlay, J. (1986) 'Automating the design of graphical presentations of relational information', *ACM Transactions on Graphics*, 5(2), pp. 110–141. <https://doi.org/10.1145/22949.22950>

Shneiderman, B. (1996) 'The eyes have it: a task by data type taxonomy for information visualizations', in *Proceedings of the 1996 IEEE Symposium on Visual Languages*. Los Alamitos, CA: IEEE Computer Society Press, pp. 336–343.